MagnaDesign

Engineering tool for PFC inductors and reactors

Internal training
Variable-frequency drive engineering | May 15, 2026

What we will cover today

App overview

Example 1 – Boost PFC 1.5 kW

Example 2 – 1ϕ line reactor, 600 W

Example 3 – Flyback DCM, 65 W

Advanced features

Status and roadmap

App overview

The problem that motivated this tool

PFC inductor engineering today involves:

- **Scattered spreadsheets** per topology
- Validation locked behind external tools (FEMM/FEMMT, specialist operator)
- Hard to iterate over L, N, core, wire without redoing everything
- EMC compliance (IEC 61000-3-2) is a separate calculation
- No **design trail** between projects

MagnaDesign integrates:

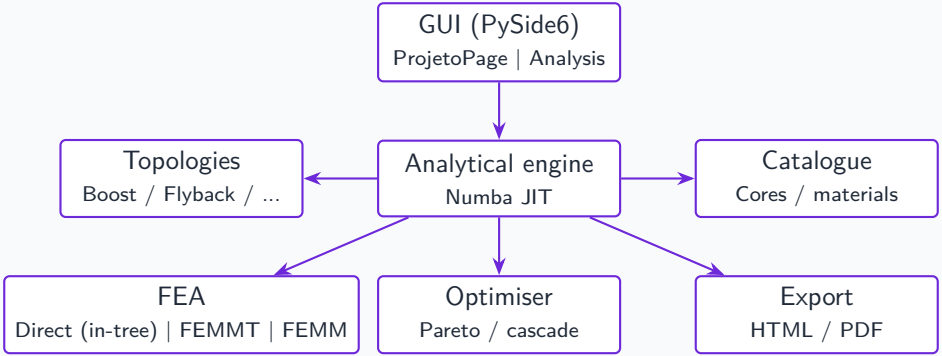
- Analytical engine, calibrated against catalogues
- **In-tree direct FEA backend** (no external install) + FEMMT and FEMM as opt-in fallbacks
- Multi-tier optimiser (cascade)
- Datasheet-style exporter (Typst PDF)
- Single GUI for 6 topologies

Supported topologies

Topology	Typical range	Applications
Boost CCM	200–2000 W	Server PSUs, industrial drives
Buck CCM	100–1500 W	Industrial DC–DC converters
Interleaved Boost	1–10 kW	High-power PFC, ripple cancel.
Flyback (DCM/CCM)	30–300 W	SMPS, adapters, auxiliaries
Line reactor ($1\phi/3\phi$)	100 W–500 kW	VFD inputs, IEC 61000-3-2
Passive choke	100 W–10 kW	1ϕ passive PFC

Each topology has its own analytical computation but shares the same engine for **losses, thermal, FEA and optimisation**.

Architecture — block view



The engine sits in the middle. Every UI surface talks to it through Pydantic models (`Spec`, `Core`, `Wire`, `Material`, `DesignResult`).

Main view — the Analysis page (top portion)

Core selection + 3D view + waveforms + losses + winding + air gap. All cards populate from the same `DesignResult` the engine emits.

Example 1 – Boost PFC 1.5 kW

Example 1 — universal-input boost PFC, 1.5 kW

Specification

- $V_{\text{in}} = 85$–$265\ \text{V}_{\text{rms}}$
- $V_{\text{out}} = 400$ V (PFC bus)
- $P_{\text{out}} = 1500$ W
- $f_{\text{sw}} = 100$ kHz
- Ripple target: 30 % of I_{pk}

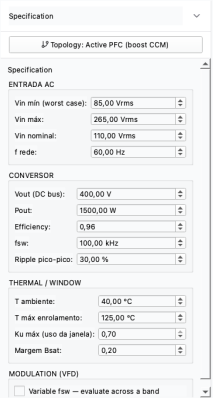
Engine output

- Magnetics 0058735A2 toroid, High-Flux 26 μ
- Round-65 heavy-build, $d_{\text{Cu}} = 3.88$ mm
- $N = 39$ turns, $L = 134\ \mu$H
- $T_{\text{w}} = 66\,^{\circ}$C, sat margin 82 %, $K_u = 0.31$

Why this example?

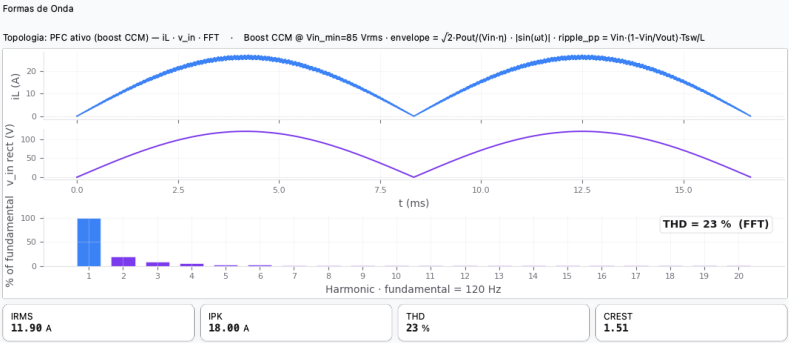
- **Most common** case in the market — server PSU, telecom, industrial drives
- Power range where litz vs solid is still a real design decision
- Demonstrates μ **rolloff** under DC bias (High-Flux 26 drops $\sim$ 40 % at peak)
- Exercises every FEA-dialog tab

1.1 — Spec drawer (user input)



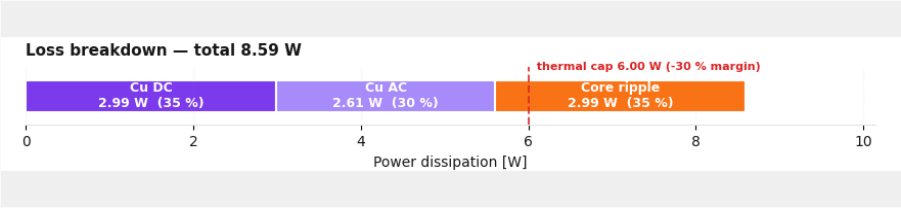
The `SpecDrawer` validates each field via Pydantic and re-runs the engine on `Enter`. Topology + V_{in} range + P_{out} + f_{sw} — three clicks and a recalc.

1.2 — Analysis: waveforms card



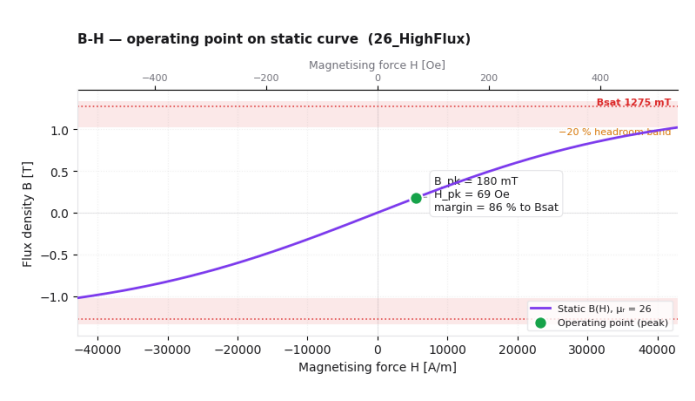
$i_L(t)$ and v_{in}(t) over the line cycle, with the harmonic FFT below and KPI tiles for I_{rms}, I_{pk}, THD and crest. THD $=$ 23 % is the inductor-only contribution; total system THD with control loop is much lower.

1.3 — Loss breakdown (proportional bar)



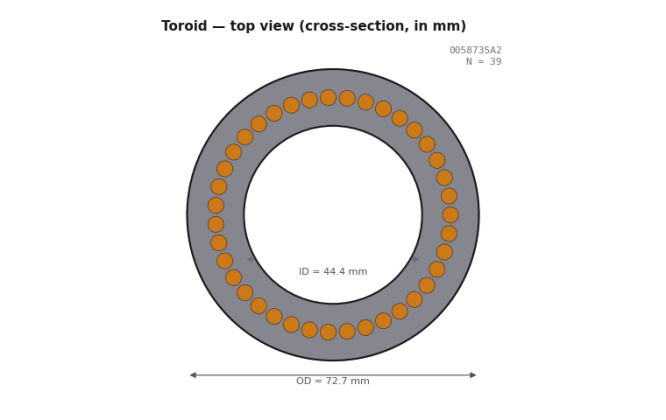
The stacked bar shows which loss family dominates the design. Cu DC concentrates 35 % of the losses — review wire gauge before chasing skin / proximity effect.

1.4 — Operating B-H curve



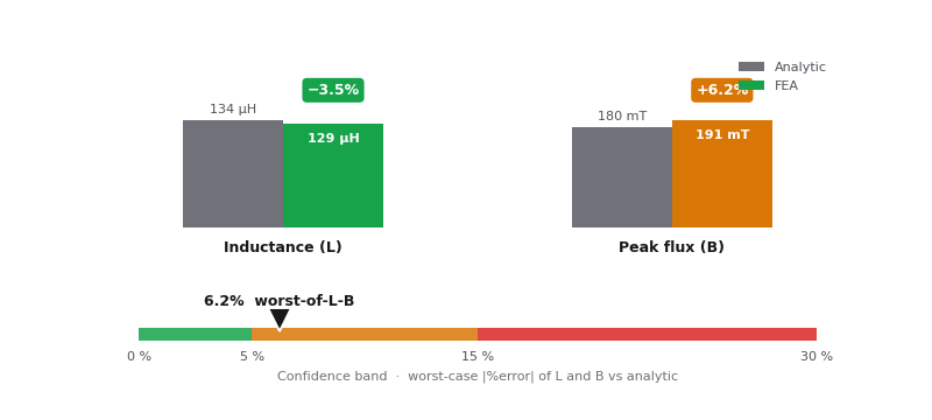
Operating-point dot (green) on the static $B(H)$ curve. Dotted lines mark B_{sat} and the $-20\,\%$ headroom band. Reported sat margin: **82 %** — conservative on purpose so the design tolerates a 50 % load surge without crossing B_{sat}.

1.5 — Geometry (datasheet-style cross section)



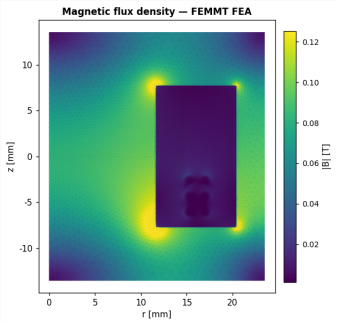
Toroid top-down view — 39 turns of round-65 wire wrapped around the donut. OD = 72.7 mm, ID = 44.4 mm, HT = 35.1 mm. Datasheet stamp: `0058735A2` • `High-Flux 26` • `N = 39`. Dimensions are derived from the catalog's $A_e/l_e/W_a$ when the vendor doesn't ship explicit OD/ID/HT.

1.6 — FEA validation: Summary tab

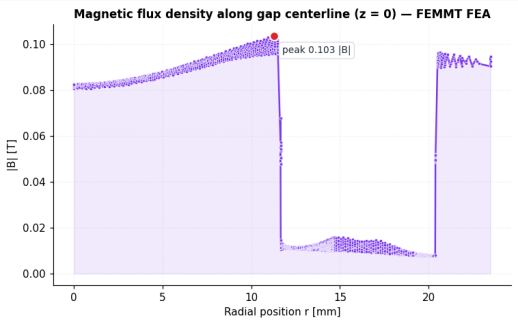


Paired bar chart *analytic vs FEA* for L and B_{pk}, with a confidence gauge below pointing at the worst of the two errors. Calibrated analytical model.

1.7 — FEA: $|B|$ heatmap + 1D centerline



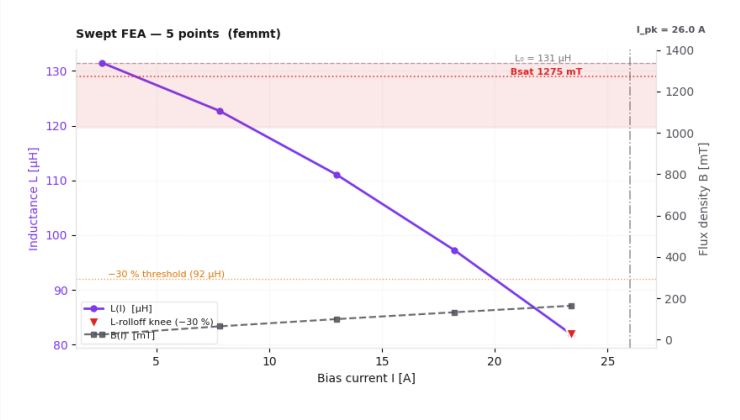
r-z, viridis



$|B|$ along z $= 0$

Click any card to open the lightbox with the per-category engineering context.

1.8 — FEA: L vs current sweep



Tier-4 swept-FEA runs the solver at N bias points. Violet line: $L(I)$. Dashed: $B(I)$ (right axis). Red band: B_{sat}. Triangle: -30% rolloff knee.

Example 2 – 1ϕ line reactor, 600 W

Example 2 — 1ϕ line-frequency reactor (VFD input)

Specification

- Mains: 230 V_{rms}, 60 Hz, 1-phase
- Load: 600 W
- $I_{rms} \approx 2.6$ A
- Target impedance: **3 %** of source
- $L \approx 5$ mH

Engine output

- Magnetics 0058340A2 toroid, High-Flux 125 μ
- Round-75 heavy-build, $d_{Cu} = 3.46$ mm
- $N = 124$ turns, $L = 5060$ μH
- $T_w = 40\,°C$, sat margin 78 %, total loss **0.27 W**
- IEC 61000-3-2 Class A

Why this example?

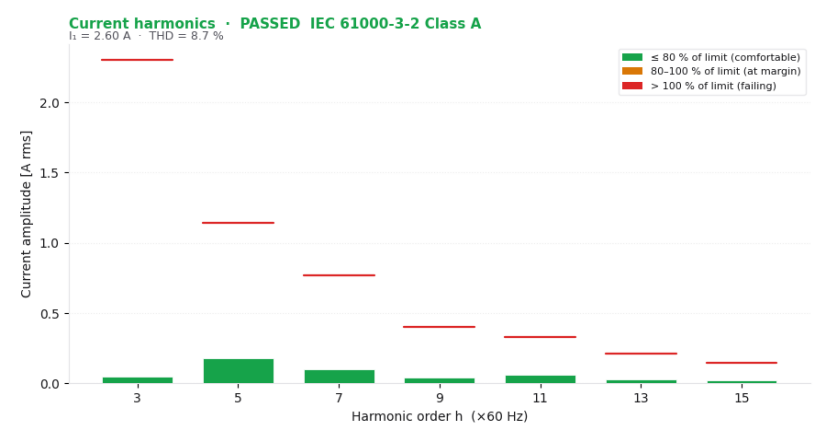
- **Line-frequency** topology (60 Hz, no high-frequency switching)
- Triggers the **IEC 61000-3-2 harmonics** card — the engine grades each order against the limit
- Powder core proves itself: $\rho_{loss} < 1$ W in a 220 g toroid
- Pass / fail EMC compliance answered visually in seconds

A 22 kW 3ϕ variant requires silicon-steel laminations (M19), which our MAS catalog does not yet ship; powder cores can't reach 22 kW at 60 Hz with sensible saturation budgets, so this slot was rescaled.

2.1 — Reactor spec + initial analysis

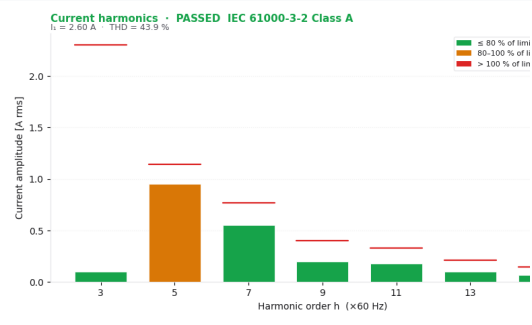
The `line_reactor` topology with `n_phases = 1` switches the inductance equation (Erickson Ch. 18) and exposes reactor-specific fields (impedance, voltage drop). Spec drawer recalculates on every `Enter`.

2.2 — Harmonics chart + IEC 61000-3-2 (Compliance tab)

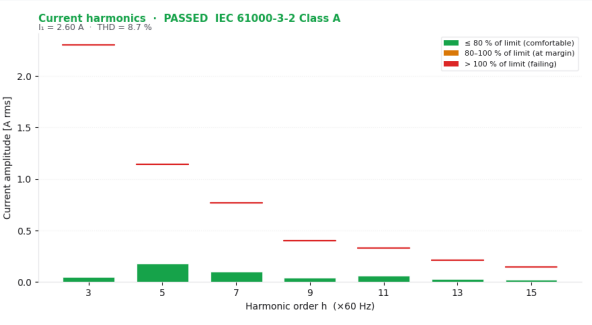


Per-order spectrum with IEC limits as red overlays. Green / amber / red coding by margin to limit; title verdict is *PASSED* or *FAILED on h = X*. The chart now lives inside the **Compliance tab**'s IEC 61000-3-2 card — spectrum visual + per-order pass/fail table + audit notes share a single panel, fed by one evaluation pass.

2.3 — Before / after the reactor



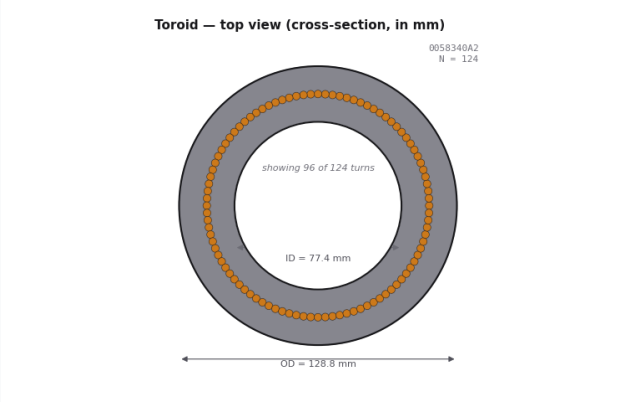
No reactor



3 % reactor

Apparent power drops $\sim 18\,\%$ for the same useful watts; THD goes from $80\,\%$ to under $10\,\%$.

2.4 — Reactor geometry (toroid view)



Top-down view of the 0058340A2 donut, OD 128.8 mm, ID 77.4 mm, HT 26.4 mm, with 124 turns of round-75 wire wrapped around the ring. The schematic painter derives the ring dimensions from $A_e/l_e/W_a$ when the catalog leaves OD/ID/HT empty.

Example 3 – Flyback DCM, 65 W

Example 3 — DCM flyback, 65 W

Specification

- $V_{\text{in}} = 85$–265 V_{rms}
- $V_{\text{out}} = 19$ V (laptop)
- $I_{\text{out}} = 3.4$ A $\Rightarrow$ 65 W
- $f_{\text{sw}} = 65$ kHz, DCM mode
- $L_{\text{p}} = 165$ μH (engine output)

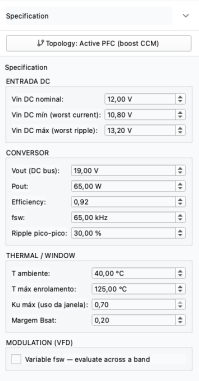
Engine output

- Thornton NPQ 32/20 ferrite (IP12E, gapped)
- Round-185 single-build, $d_{\text{Cu}} = 0.97$ mm
- $N_{\text{p}} = 26$, $N_{\text{s}} = 6$ ($n = 4.0$)
- $T_{\text{w}} = 54\,^{\circ}$C, sat margin 41 %, $K_u = 0.26$

Why this example?

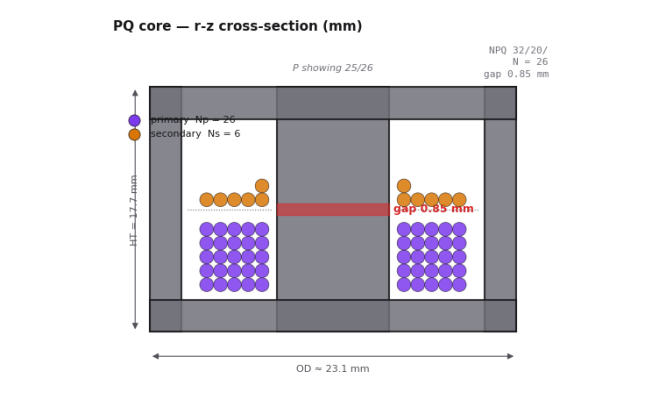
- Coupled-pair — entirely different from a single-winding choke
- Triggers the **2-winding render** (primary + secondary in distinct colours) on the geometry view
- Exercises the **leakage** L_{lk} calculation (Mohan / Erickson formula)
- DCM — discontinuous triangular current waveform

3.1 — Flyback spec



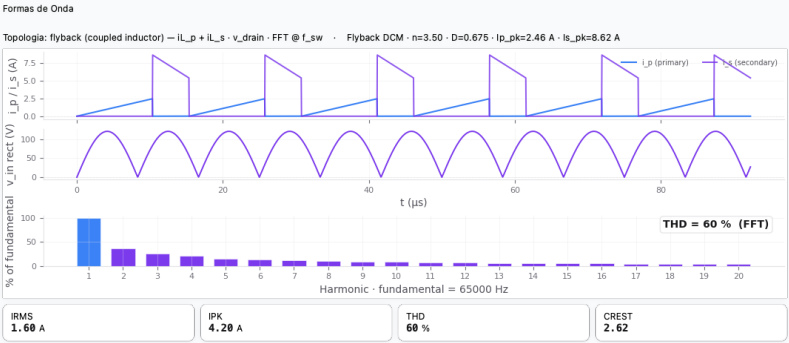
Extra fields vs. boost: `Vout`, `n_turns ratio`, CCM/DCM mode inferred when $D < 0.5$.

3.2 — Geometry: coupled-pair cross section



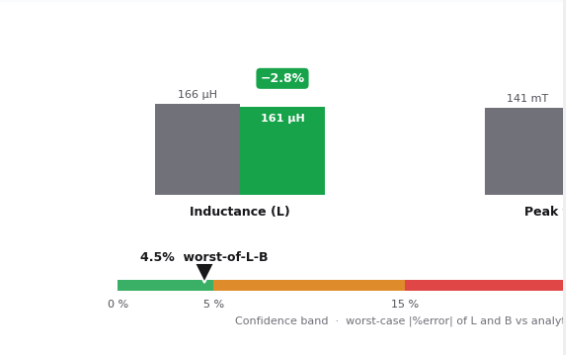
NPQ 32/20 with **primary in violet** (26 turns, bottom of each window) and **secondary in amber** (6 turns, top). Centre-leg air gap of 1.5 mm shown in red. Both windows are mirrored — a real bobbin uses a single window split horizontally.

3.3 — Flyback waveforms + losses

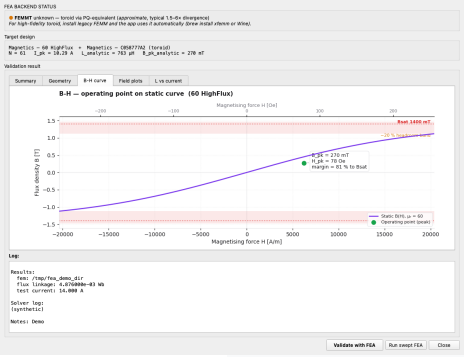


Primary $i_p(t)$: triangular DCM with a visible dead-time. The reflected secondary current $i_s = n \cdot i_p$ peaks $4\times$ higher. L_{lk} estimated via Mohan's window-split formula. $V_{DS,pk} = V_{in,pk} + nV_{out} \approx 450\,\text{V} \Rightarrow$ a 600 V FET still fits.

3.4 — Flyback FEA validation



Summary tab

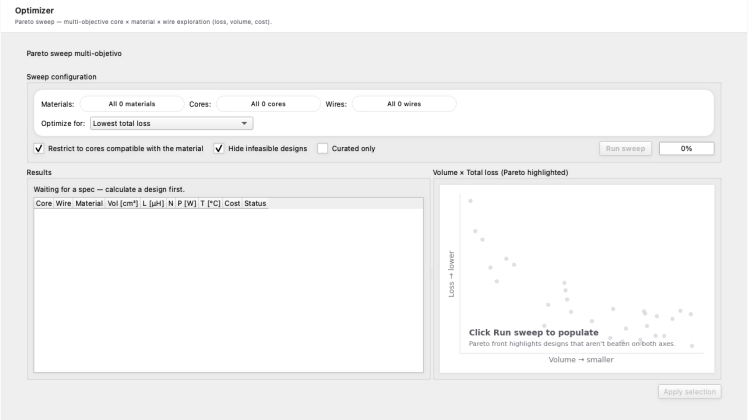


Operating B-H

For coupled inductors the FEA validates the primary L_{p}; coupling k is then derived analytically from $L_{\mathrm{lk}}/L_{\mathrm{p}}$.

Advanced features

Pareto optimiser (single-tier)



Cartesian sweep over (`core`, `material`, `wire`, `N_turns`) — generates 1 000–10 000 candidates.

Dominated points faded; the front in violet; the user's pick highlighted as a star.

Cascade optimiser (multi-tier)

- **Tier 1** — pure analytical (seconds)
- **Tier 2** — full engine with skin / proximity (minutes)
- **Tier 3** — single-point FEA at peak ($\sim$ 30 s/candidate)
- **Tier 4** — swept-FEA $L(I)$ with rolloff (several minutes/candidate)

The Top-N table shows each tier's refined values. A virtual COALESCE column always shows the most-recent number for every row.

Direct FEA backend — in-tree replacement

Why we built our own

- FEMMT covered only 6/12 catalogue shapes (no RM / P / EP / EFD / EI, no toroid powder)
- 12 s per solve ⇒ optimisation cascades took minutes
- Required ONELAB download + first-run install dialog on Windows
- Crashes on common geometries (high-N toroid, EI ferrite cuts)

What direct gives us (v0.5.0+)

- **12/12** catalogue shapes
- **0.3 s** median solve (∼40× faster)
- **Zero external install** — pure Python, no ONELAB, no FEMMT pip wheel
- Reluctance + Roters fringing + toroidal closed-form + Dowell AC + lumped thermal
- 13-file engineering documentation set under `fea/direct/docs/`

Engine ↔ backend parity — the contract

Case	|L %| before	after	Root cause / fix
Si-Fe line reactor (EI3311 30 A)	174 %	0.00 %	Closed-path gate
Toroid ferrite (T107/65/18)	1577 %	2.1 %	Closed-shape gate
Ferrite boost PQ40/40	120 %	0.3 %	Roters fringing iter.
Ferrite boost ETD49	96 %	0.7 %	Roters fringing iter.
Median (29 boost-PFC cases)	**166 %**	**0.22 %**	*combined*

Two analytical solvers run side-by-side on every recalc: the design-loop analytical engine and the direct FEA backend. They must agree to numerical precision when fed the same inputs — a hard parity contract enforced by 8 regression tests including bit-for-bit Roters factor comparisons between the two duplicate implementations. Pre-fix users saw a "$+30\,\%$ between FEA and analytic" warning; with the closed-path gates + iterative fringing-aware gap sizing, that disagreement is now noise.

Modulation bands — fsw and load

Variable fsw (compressor VFD)

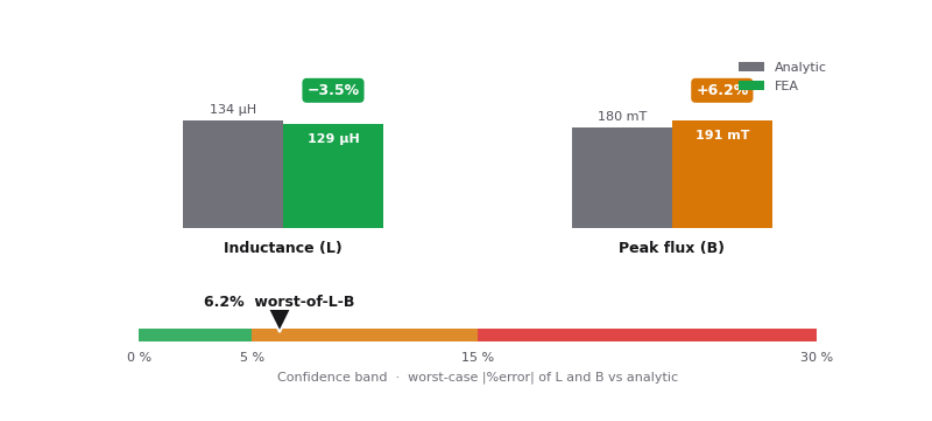
- Switching freq. sweeps a band as the compressor RPM varies
- Profiles: `uniform`, `triangular_dither`, `rpm_band` (derives fsw from RPM $\times$ pole pairs)
- Engine evaluates N points across the band, returns per-metric worst-case

Variable load (Pout sweep, new in v0.5.6)

- Sibling of fsw band for the load dimension
- Profiles: `uniform`, `triangular_dither`, `compressor_swing` (IEC 60335 50–130 % range)
- Mutually exclusive with fsw band — 2-D sweep would explode evaluation cost

Both bands surface as a 3-up envelope chart in the Analysis tab ($P_{total} \rightarrow B_{pk} \rightarrow \Delta T$). The chart widget is axis-agnostic — it reads each `BandPoint.swept_value()` and labels the X axis as `Pout [W]` or `fsw [kHz]` based on which band the spec is sweeping.

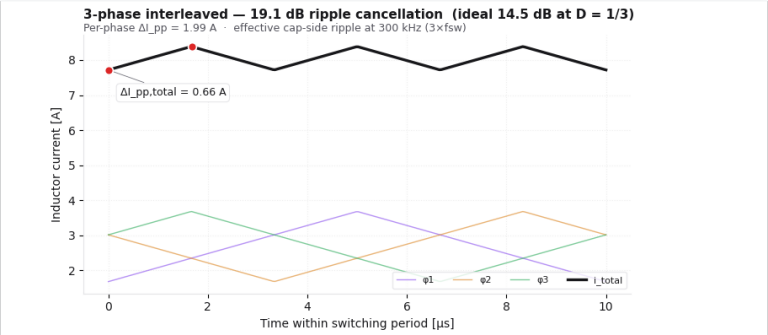
FEA validation dialog — 5 tabs



Five tabs share one solver result: **Summary** (analytic vs FEA bars + confidence gauge) | **Geometry** (datasheet cross-section) | **B-H** (operating point on static curve) | **Field plots** (heatmap + 1D centerline + histogram) | **L vs current** swept. Backed by the direct in-tree solver — no external install required.

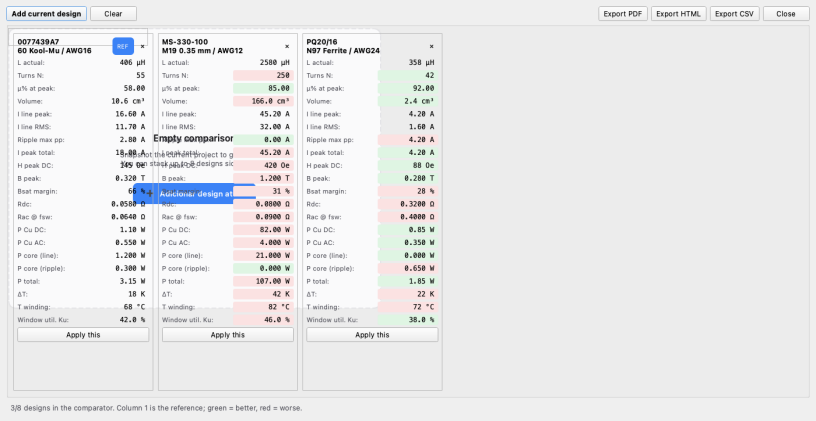
Topology-specific cards

Phase ripple — interleaved cancellation N-phase



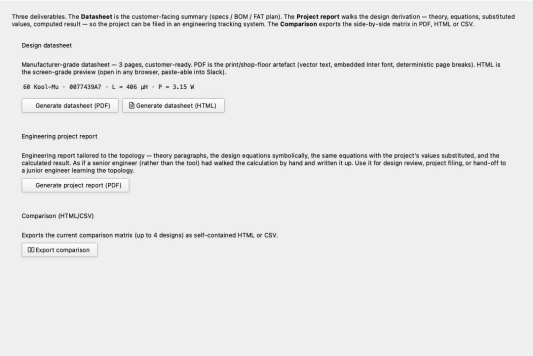
Cards self-hide via `setVisible(False)` when the topology doesn't apply — the row collapses and the user only sees what's relevant to their design. Example shown: *Phase overlay* (interleaved boost). The IEC 61000-3-2 spectrum view used to live here too; it moved to the Compliance tab so the chart sits next to the per-order pass/fail table.

Compare designs (side by side)



Diff table across 2–3 `DesignResults`. Different values per metric highlighted in violet. Useful to justify a core / material swap in a design review.

Export — HTML / PDF datasheet



- TDK / Würth / Vishay layout
- Front / top / side / iso views
- Full electrical-spec table
- Topology-specific performance section
- HTML self-contained (offline) or PDF

3D viewer

PyQt + Qt3D render of the toroid / E-core with winding visualised. Useful to confirm the assembly before sending to the bobbin manufacturer.

Coupled thermal model

Default path — direct backend (lumped)

- Single-node Churchill-Chu convection model inside `fea/direct/physics/thermal.py`
- One click after the magnetostatic pass: *Run thermal* — no external solver, <1 ms per call
- Material-keyed conductivity defaults: ferrite 5, silicon-steel 25, powder 20, nanocrystalline 10, amorphous 12 W/(m K
- Iterative EM-thermal coupling (`em_thermal_coupling.py`) picks up $R(T) \to F_R(T) \to P_{cu}(T)$ feedback in 3–8 iterations

Optional path — FEMMT thermal (legacy)

Default outcome (direct)

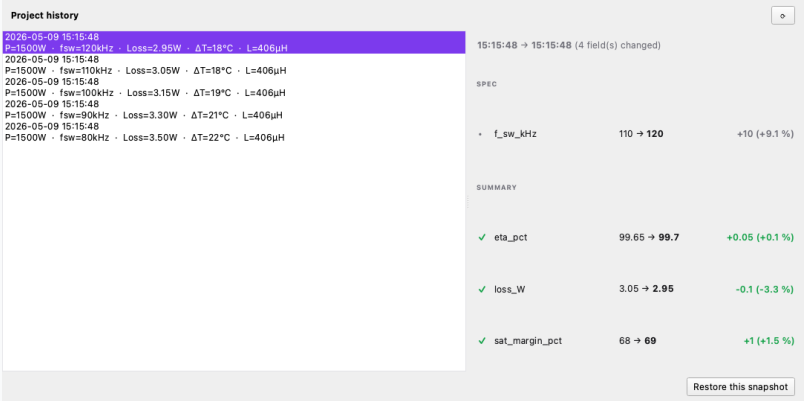
The dialog logs:

- `T_winding` (volume-avg)
- ΔT vs ambient
- Verdict chip: *passes* or *over 105 °C* (class-B insulation budget)

Trade-off

Lumped 1-node returns volume-average T; hot-spot on tall multi-layer windings can be 5–15 K higher. For tight thermal designs the iterative coupling closes the gap to ∼2 K vs camera measurement.

Project history — git-like timeline



Every recalc is a snapshot. The History tab shows the timeline (newest first); clicking any row diffs it against the immediately-prior snapshot, with **green** for changes that improved a metric (lower loss / lower ΔT / higher η) and **red** for worsens. *Restore this snapshot* rebuilds the spec drawer + selection from the chosen point and triggers a recalc — history stays append-only.

Status and roadmap

Where we are today (v0.5.7)

Shipped & validated

- 6 topologies with analytical engine
- **Direct in-tree FEA backend** (12/12 shapes, 0.3 s per solve)
- FEMMT + FEMM available as opt-in legacy
- Single + cascade (4-tier) optimiser
- Topology-specific cards (auto hide/show)
- Typst-based PDF datasheet exporter
- 5-tab FEA validation dialog
- Iterative EM-thermal coupling
- **Project history** — git-like timeline + visual diffs

Performance

- Engine: **3.6× speedup** via fused Numba kernel
- Tier-1 analytical: ∼ 1 ms
- Direct FEA solve: **0.3 s** (40× faster than FEMMT)
- Tier-4 swept (5 pts): ∼2 s with direct
- Sparkle-style auto-update + Ed25519 sig

Engine ↔ FEA parity

- Median $|L\,\%|$: 166 % → 0.22 % after Si-Fe / Roters / toroid fixes
- 26/29 boost-PFC cases within 5 %
- 8 regression tests pin the contract

Roadmap — next releases

- **Variable Vin modulation** — sibling of fsw + load bands for brown-out / universal-input envelopes
- **Voltage stress** per component (FET, diode)
- **Auto-updating magnetics catalogue** via the vendors' APIs (Magnetics, Micrometals, TDK)
- **LP-series catalogue audit** — Magnetics LP cores still show an AL-convention mismatch (engine vs direct disagree by 87×)
- **3-D FEM** (deferred Phase 4.2) — tet mesh for cases where the Roters clamp at $k = 3$ becomes the bottleneck
- **Multi-design report bundle** — consolidated PDF for an entire bill-of-materials

Questions?

MagnaDesign— May 15, 2026

Bibliographic references

- Erickson & Maksimović, *Fundamentals of Power Electronics*, 3rd ed., Ch. 18 (PFC).
- Mohan, *Power Electronics*, 3rd ed., Ch. 4 (flyback leakage).
- Hwu & Yau, *Resonance reduction in interleaved boost PFC*, IEEE TPE 2009.
- IEC 61000-3-2 Ed. 5 — Harmonic emission limits.
- Magnetics Inc., *Powder cores design manual* (Kool-Mu, High-Flux).
- Würth Elektronik, *Trilogy of Magnetics*, 5th ed.